

EFGP log-marginal gradient: derivation and algorithm

1 Log-marginal gradient

Let $\tilde{K} := K_\theta + \sigma^2 I$ with $K_\theta \in \mathbb{R}^{N \times N}$. For any hyper θ ,

$$\log p(y \mid \theta) = -\frac{1}{2} y^\top \tilde{K}^{-1} y - \frac{1}{2} \log \det \tilde{K} + \text{const}, \quad (1)$$

$$\partial_\theta \log p = \frac{1}{2} (T_2 - T_1), \quad T_1(\theta) = \text{tr}(\tilde{K}^{-1} \partial_\theta \tilde{K}), \quad T_2(\theta) = y^\top \tilde{K}^{-1} \partial_\theta \tilde{K} \tilde{K}^{-1} y. \quad (2)$$

Let $\alpha := \tilde{K}^{-1} y$. Then $T_2(\theta) = \alpha^* \partial_\theta \tilde{K} \alpha$.

2 EFGP approximation

Pick an equispaced frequency grid $\{\xi_k\}_{k=1}^M \subset \mathbb{R}^d$ of spacing h . Define

$$F \in \mathbb{C}^{N \times M}, \quad F_{j,k} = e^{i2\pi x_j \cdot \xi_k}, \quad D = \text{diag}(\sqrt{h^d \hat{k}_\theta(\xi_k)}), \quad \Phi := FD. \quad (3)$$

Then

$$K_\theta \approx \Phi \Phi^* = F D^2 F^*, \quad \partial_\theta K_\theta \approx F D'_\theta F^*, \quad D'_\theta := \text{diag}(h^d \partial_\theta \hat{k}_\theta(\xi_k)). \quad (4)$$

Noise is outside the $FD'F^*$ family: $\partial_{\sigma^2} \tilde{K} = I$.

Two $M \times M$ operators are available at $\mathcal{O}(M \log M)$:

$$T := F^* F \text{ (multi-level Toeplitz)}, \quad A_{\text{mean}} := \sigma^2 I + DTD = \sigma^2 I + \Phi^* \Phi. \quad (5)$$

3 Mean problem and $F^* \alpha$

Define $\beta_0 := A_{\text{mean}}^{-1} DF^* y$. Sherman–Morrison–Woodbury on $\tilde{K} = \sigma^2 I + \Phi \Phi^*$ gives

$$\alpha = \tilde{K}^{-1} y = \sigma^{-2} (y - \Phi \beta_0) = \sigma^{-2} (y - FD \beta_0), \quad (6)$$

and applying F^* ,

$$F^* \alpha = \sigma^{-2} (F^* y - T(D \beta_0)). \quad (7)$$

Key identity. The defining equation $A_{\text{mean}} \beta_0 = DF^* y$ expands as $\sigma^2 \beta_0 + DTD \beta_0 = DF^* y$, hence

$$D(F^* y - T(D \beta_0)) = \sigma^2 \beta_0 \implies \boxed{D F^* \alpha = \beta_0} \quad (\text{at CG convergence}). \quad (8)$$

Equivalently, $F^* \alpha = \beta_0 / D$ entrywise on $\text{supp}(D)$.

4 T_2 : all hypers in M -space

Kernel hypers. Using $\partial_\theta K \approx F D'_\theta F^*$,

$$T_2(\theta) = \alpha^* (F D'_\theta F^*) \alpha = \langle F^* \alpha, D'_\theta F^* \alpha \rangle \stackrel{(8)}{=} \boxed{\langle \beta_0, (D'_\theta / |D|^2) \beta_0 \rangle}. \quad (9)$$

The right form uses only β_0 (the CG output), no $F^* \alpha$. Under *inexact* CG with residual $r := DF^* y - A_{\text{mean}} \beta_0$, eq. (7) gives $DF^* \alpha = \beta_0 + r/\sigma^2$, so the $F^* \alpha$ -form of T_2 carries error $\mathcal{O}(\|r\|/\sigma^2)$ and $\mathcal{O}(\|r\|^2/\sigma^4)$. The β_0 -form's error is linear in the solution error $\|\beta_0 - A_{\text{mean}}^{-1} DF^* y\|$, with no $1/\sigma$ amplification.

Variance σ_f^2 . For SE/Matérn, $\hat{k}$ is linear in σ_f^2 , so $D^2 = \sigma_f^2 \cdot (\text{stuff})$, giving $D'_{\sigma_f^2} = D^2/\sigma_f^2$ and

$$T_2(\sigma_f^2) = \|\beta_0\|^2/\sigma_f^2 \quad (\text{special case of (9)}). \quad (10)$$

Noise σ^2 . Since $\partial_{\sigma^2} \tilde{K} = I$, $T_2(\sigma^2) = \|\alpha\|^2$. Expand directly from $\sigma^2 \alpha = y - FD\beta_0$:

$$\boxed{\|\alpha\|^2 = \sigma^{-4} (\|y\|^2 - 2\Re\{\beta_0^* DF^* y\} + \beta_0^* DTD \beta_0)}. \quad (11)$$

This is the literal expansion of $\|y - FD\beta_0\|^2/\sigma^4$; it does *not* invoke $A_{\text{mean}}\beta_0 = DF^* y$, so it is CG-robust. All three scalars are M -dots on already-computed vectors ($F^* y$, $D\beta_0$, $T(D\beta_0)$).

5 T_1 : trace in M -space

Kernel hypers. By cyclicity,

$$T_1(\theta) = \text{tr}(\tilde{K}^{-1} F D'_\theta F^*) = \text{tr}(B D'_\theta), \quad B := F^* \tilde{K}^{-1} F \in \mathbb{C}^{M \times M}. \quad (12)$$

Woodbury on $\tilde{K}^{-1}$:

$$B = \sigma^{-2} (T - TD A_{\text{mean}}^{-1} DT). \quad (13)$$

For Rademacher probes $V \in \{\pm 1\}^{T_p \times M}$, let $u_k^\theta := D'_\theta v_k$ and $X_k^\theta := A_{\text{mean}}^{-1} DT u_k^\theta$. Then $B D'_\theta v_k = \sigma^{-2} (T u_k^\theta - TD X_k^\theta)$.

β -form identity. The defining equation $A_{\text{mean}} X_k^\theta = DT u_k^\theta$ expands as $\sigma^2 X_k^\theta + DTD X_k^\theta = DT u_k^\theta$, hence

$$D(T u_k^\theta - TD X_k^\theta) = \sigma^2 X_k^\theta \implies \boxed{B D'_\theta v_k = X_k^\theta / D} \quad (\text{at CG convergence}). \quad (14)$$

So the Hutchinson estimator reduces to

$$T_1(\theta) \approx \frac{1}{T_p} \sum_k \langle v_k, X_k^\theta / D \rangle, \quad (15)$$

using only the CG output X_k^θ — no σ^{-2} subtraction. By the same argument as for T_2 (§4), this swaps the $1/\sigma^2$ amplification of the residual-form for a simple division by D . Each probe costs 1 solve against A_{mean} ; **no Toeplitz beyond the rhs setup, no NUFFT**.

Noise. $\text{tr}(\tilde{K}^{-1}) = \sigma^{-2}(N - \text{tr}(A_{\text{mean}}^{-1} DTD))$, estimated by Hutchinson on $A_{\text{mean}}^{-1} DTD$ (same batched CG).

Variance (closed form). $K = \sigma_f^2 \cdot (\partial_{\sigma_f^2} K)$, so

$$T_1(\sigma_f^2) = \frac{1}{\sigma_f^2} \text{tr}(\tilde{K}^{-1} K) = \frac{N - \sigma^2 T_1(\sigma^2)}{\sigma_f^2}. \quad (16)$$

No separate Hutchinson call.

6 Algorithm

Let $\theta_1, \dots, \theta_{K-1}$ be the kernel hypers excluding σ_f^2 . Setup: form v_{ker} for T , D , all $\{D'_\theta\}$ via `spectral_grad`, and a Jacobi preconditioner for A_{mean} .

Algorithm 1 EFGP log-marginal gradient: one NUFFT, two CG calls.

- 1: $z \leftarrow F^* y$ ▷ **only NUFFT**
 - 2: Solve $A_{\text{mean}} \beta_0 = Dz$ by CG. (**CG #1, mean**)
 - 3: $w \leftarrow T(D\beta_0)$ ▷ Toeplitz mat-vec, reused below
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- 4: *Term T_2 (all M -space):*
 - 5: kernel hypers: $T_2(\theta_i) \leftarrow \langle \beta_0, (D'_{\theta_i}/|D|^2) \beta_0 \rangle$
 - 6: variance: $T_2(\sigma_f^2) \leftarrow \|\beta_0\|^2 / \sigma_f^2$
 - 7: noise: $T_2(\sigma^2) \leftarrow \sigma^{-4} (\|y\|^2 - 2\Re\{\beta_0^* Dz\} + \beta_0^* Dw)$
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- 8: *Term T_1 :* draw $V \in \{\pm 1\}^{T_p \times M}$, $V_n \in \{\pm 1\}^{T_p \times M}$.
 - 9: for each θ_i : $\text{rhs}^i \leftarrow DT(D'_{\theta_i} V)$
 - 10: noise: $\text{rhs}^n \leftarrow DT(DV_n)$
 - 11: stack all RHSs; solve $A_{\text{mean}} X = \text{rhs}$ by batched CG. (**CG #2, trace**)
 - 12: kernel (β -form): $T_1(\theta_i) \leftarrow \text{mean}_k \langle (V)_k, (X^i)_k / D \rangle$
 - 13: noise: $T_1(\sigma^2) \leftarrow \sigma^{-2} (N - \text{mean}_k \langle (V_n)_k, (X^n)_k \rangle)$
 - 14: variance: $T_1(\sigma_f^2) \leftarrow (N - \sigma^2 T_1(\sigma^2)) / \sigma_f^2$
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- 15: **Return** $\nabla_\theta \log p = \frac{1}{2}(T_2 - T_1)$ for each hyper.
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Complexity. Per gradient: one NUFFT ($\mathcal{O}(N + M \log M)$), two CG calls against A_{mean} (mean: one RHS; trace: $K T_p$ RHSs), all $\mathcal{O}(M \log M)$ per mat-vec. Term T_2 is $\mathcal{O}(KM)$ total. No NUFFT inside the gradient loop.

Why the β -form everywhere. Both T_2 and T_1 have the same structural choice: either take the CG output and divide by D (the β -form, eq. (8), (14)), or Woodbury-reconstruct via $(F^* y - T(D\beta_0)) / \sigma^2$ (for T_2) or $(Tu - TDX) / \sigma^2$ (for T_1). Both are algebraically identical at CG convergence. Under inexact CG with residual δ , the Woodbury-reconstructed form carries a $1/\sigma^2$ prefactor that amplifies δ . In T_2 this pre-amplified quantity sits on *both sides* of a quadratic form, giving a $\|\delta\|^2 / \sigma^4$ term that can flip the gradient sign. In T_1 it sits on one side of a linear inner product, giving only a $\|\delta\| / \sigma^2$ bias. The β -form avoids the $1/\sigma$ amplification in both cases; its residual scales by $1/D$ instead, which is bounded on the EFGP frequency grid.